Statistical Estimators

Estimators

An *estimator* is an operation computing the value of an estimate, that targets the parameter, using measurements from a sample.

Estimator Definition

Given a *random sample*, $X = (X_1, \dots, X_n)^{\mathrm{T}}$ from a distribution $F(\theta)$, an estimator is defined as

sample

$$\hat{\theta} = h(X)$$

$$\hat{\theta} \to \theta$$

Example Estimator

$\hat{\theta}_1 = X, \quad \hat{\theta}_2 = \frac{1}{n}\sum X_i, \quad \hat{\theta}_3 = \prod_{i=1}^{n} X_i$

$\hat{\theta}_4 = X_{[n/2]} \leftarrow$ 50th percentile

Given a *random sample*, $X = (X_1, \ldots, X_n)^{\mathrm{T}}$ from a distribution $F(\theta)$

Methods of Finding Estimators

- Maximum Likelihood Estimator
- Method of Moments Estimator
- Bayesian Estimtors

Evaluating Estimators

- Mean Squared Error
- Unbiased Estimator
- Sufficiency

Unbiased Estimator

An unbiased estimator $\widehat{\theta}$ is an estimator that satisfies the following condition:

$$E(\widehat{\theta}) = \theta$$

Bias

The bias of an estimator is defined as

$$B(\hat{\theta}) = E(\hat{\theta}) - \theta$$

Mean Square Error

The mean square error of an estimator is $\hat{\theta}$ is given as

$$\begin{aligned} MSE(\hat{\theta}) &= E\{(\hat{\theta} - \theta)^2\} \\ &= Var(\hat{\theta}) + B(\hat{\theta})^2 \end{aligned}$$

Is $\bar{X}$ an unbiased estimator of μ?

Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, find the bias of $\bar{X}$.

$E(X_i) = \mu \quad Var(X_i) = \sigma^2$

$E(X_i^2) = \sigma^2 + \mu^2$

$Var(X_i) = E(X_i^2) - E(X_i)^2$

$E(\bar{x}) - \mu$

$$E(\bar{x}) = E\left(\frac{1}{n}\sum x_i\right)$$

$$= \frac{1}{n} E\left(\sum x_i\right) = \frac{1}{n}\sum E(x_i)$$

$$\frac{1}{n}\sum \mu = \frac{\cancel{n}\mu}{\cancel{n}}$$

$$\mu - \mu = 0$$

$\bar{x}$ is unbiased for μ

Why is S^2 divided by $n-1$ instead of n?

Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, find the bias of S^2.

$E(X_i) = \mu \qquad E(X_i^2) = \sigma^2 + \mu^2$

$\bar{X} \sim N(\mu, \sigma^2/n)$

$E(\bar{X}) = \mu \qquad Var(\bar{X}) = \sigma^2/n \qquad E(\bar{X}^2) = \frac{\sigma^2}{n} + \mu^2$

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2 \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2$$

$$E(\hat{\sigma}^2) - \sigma^2 \stackrel{?}{=} 0$$

$$E(\hat{\sigma}^2) = E\left(\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2\right)$$

$$= \frac{1}{n}E\left(\sum(x_i^2 - 2x_i\bar{x} + \bar{x}^2)\right)$$

$$= \frac{1}{n}E\left(\sum x_i^2 - \sum 2x_i\bar{x} + \sum \bar{x}^2\right)$$

$$= \frac{1}{n}E\left(\sum x_i^2 - 2\bar{x}\sum x_i + n\bar{x}^2\right)$$

$$2\bar{x}\,n\bar{x}$$

$$= \frac{1}{n} E\left(\sum x_i^2 - 2n\bar{x}^2 + n\bar{x}^2\right)$$

$$= \frac{1}{n} E\left(\sum x_i^2 - n\bar{x}^2\right)$$

$$= \frac{1}{n}\left[E\left(\sum x_i^2\right) - n E\left(\bar{x}^2\right)\right]$$

$$\frac{1}{n}\left[\sum E\left(x_i^2\right) - n E\left(\bar{x}^2\right)\right]$$

$$\frac{1}{n}\left[\sum \underline{\left(\sigma^2 + \mu^2\right)} - n\left(\frac{\sigma^2}{n} + \mu^2\right)\right]$$

$$\frac{1}{n}\left[n\sigma^2 + \cancel{n\mu^2} - \sigma^2 - \cancel{n\mu^2}\right]$$

$$\frac{n\sigma^2 - \sigma^2}{n} = \sigma^2\left(\frac{n-1}{n}\right)$$

$$E(\hat{\sigma}^2) = \sigma^2\left(\frac{n-1}{n}\right)$$

$$\sigma^2\left(\frac{n-1}{n}\right) - \sigma^2 = \sigma^2\left(\frac{n-1}{n} - \frac{n}{n}\right)$$

$$\sigma^2\left(\frac{-1}{n}\right)$$

$$\left(\frac{n}{n-1}\right)\hat{\sigma}^2 = \frac{\cancel{n}}{n-1} \cdot \frac{1}{\cancel{n}} \sum (x_i - \bar{x})^2$$

$$= \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

$$E\left(\frac{n}{n-1}\hat{\sigma}^2\right) = \frac{n}{n-1} \underbrace{E(\hat{\sigma}^2)}_{\sigma^2\left(\frac{n-1}{n}\right)} = \frac{\cancel{n}}{\cancel{n-1}} \sigma^2 \frac{\cancel{n-1}}{\cancel{n}}$$

$$= \sigma^2$$

Problem Functions of RV

Let X_1, X_2, X_3 follow and exponential distribution with mean and variance λ and λ^2, respectively. Using the following estimators:

$\text{Bias}(\hat{\theta}_i)$ $E(\hat{\theta}) - \lambda$

- $\hat{\theta}_1 = X_1$ $\quad 0 \quad Var(\hat{\theta}_1) = \lambda^2$
- $\hat{\theta}_2 = \frac{X_1+X_2}{2}$ $\quad 0 \quad Var(\hat{\theta}_2) = \frac{\lambda^2}{2}$
- $\hat{\theta}_3 = \frac{X_1+2X_2}{3}$ $\quad 0 \quad Var(\hat{\theta}_3) = 5/9\,\lambda^2$
- $\hat{\theta}_4 = \frac{X_1+X_2+X_3}{3}$ $\quad 0 \quad Var(\hat{\theta}_4) = \frac{\lambda^2}{3}$

Identify which estimator: $MSE(\hat{\theta}) = Var(\hat{\theta}) - Bias(\hat{\theta})^2$

1. Is unbiased?

Likelihood Function

Using the joint pdf or pmf of the sample X, the likelihood function is a function of θ, given the observed data $X = x$, defined as

joint Dist.

$$L(\theta|x) = f(x|\theta)$$

If the data is iid, then

$L(\theta|x) \sim$ Prob of observing the data

$$f(x|\theta) = \prod_{i=1}^{n} f(x_i|\theta)$$

$$\frac{dF(x|\theta)}{d\theta} = \frac{d}{d\theta} \prod F(x_i|\theta)$$

Log-Likelihood Function

If $\ln\{L(\theta)\}$ is monotone of θ, then maximizing $\ell(\theta) = \ln\{L(\theta)\}$ will yield the maximum likelihood estimators.

Binomial Distribution

Let $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Bin}(n, p)$

Poisson Distribution

Let $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Pois}(\lambda)$

Normal Distribution

Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$

Exponential Distribution

Let $X_1, \dots, X_n \overset{iid}{\sim} Exp(\lambda)$